

COM3502-4502-6502 SPEECH PROCESSING

Lecture 18

Linear Prediction



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Linear Prediction

'Linear prediction' is the basis for almost all modern speech coding algorithms ...

- mobile phones
- internet telephony (VOIP)
- military communications



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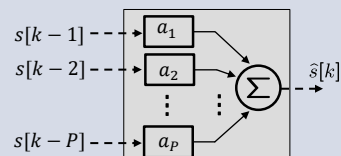
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Linear Prediction Analysis

- Linear prediction assumes that each sample (of a speech signal) can be predicted from a weighted sum of the P preceding samples

$$\hat{s}[k] = a_1 s[k-1] + a_2 s[k-2] + \dots + a_P s[k-P]$$

$$= \sum_{i=1}^P a_i s[k-i]$$

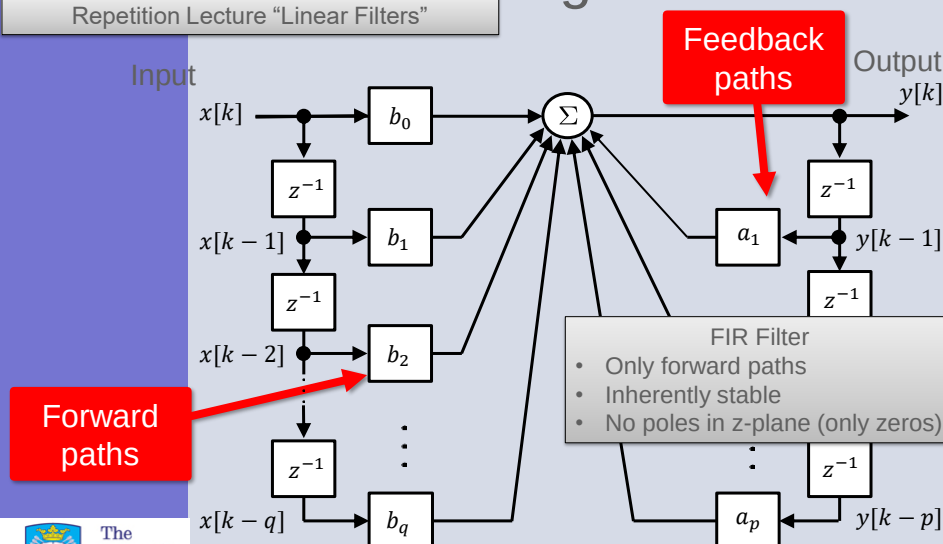


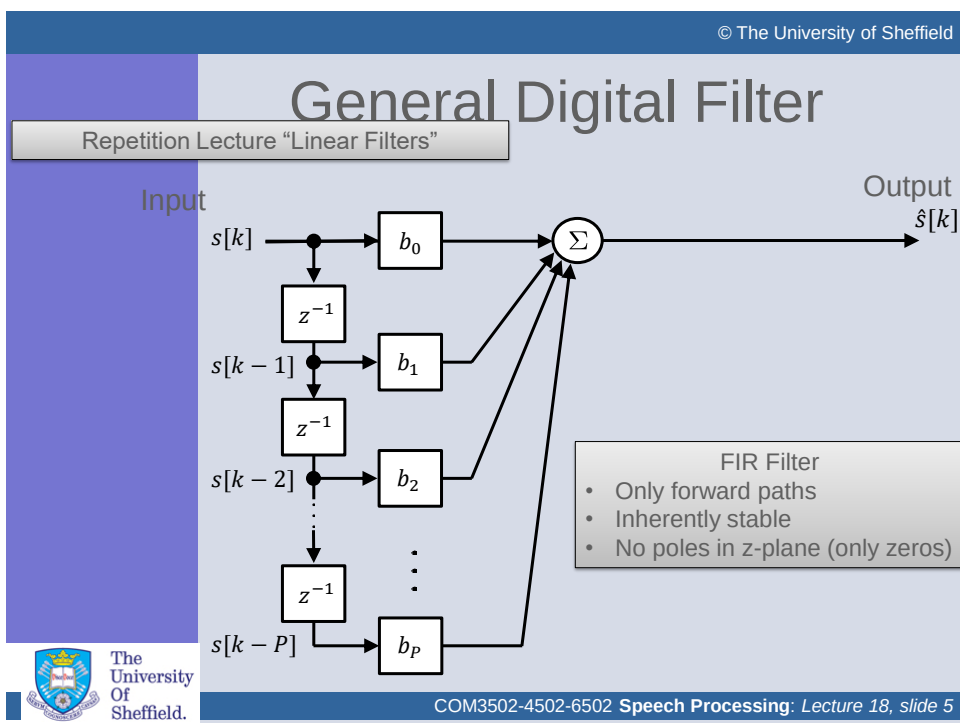
- Exploiting redundancy arising from the *stationarity* assumption, linear prediction allows a speech frame of several hundred samples to be represented by only 10-15 'prediction coefficients' ($a_1 \dots a_P$)



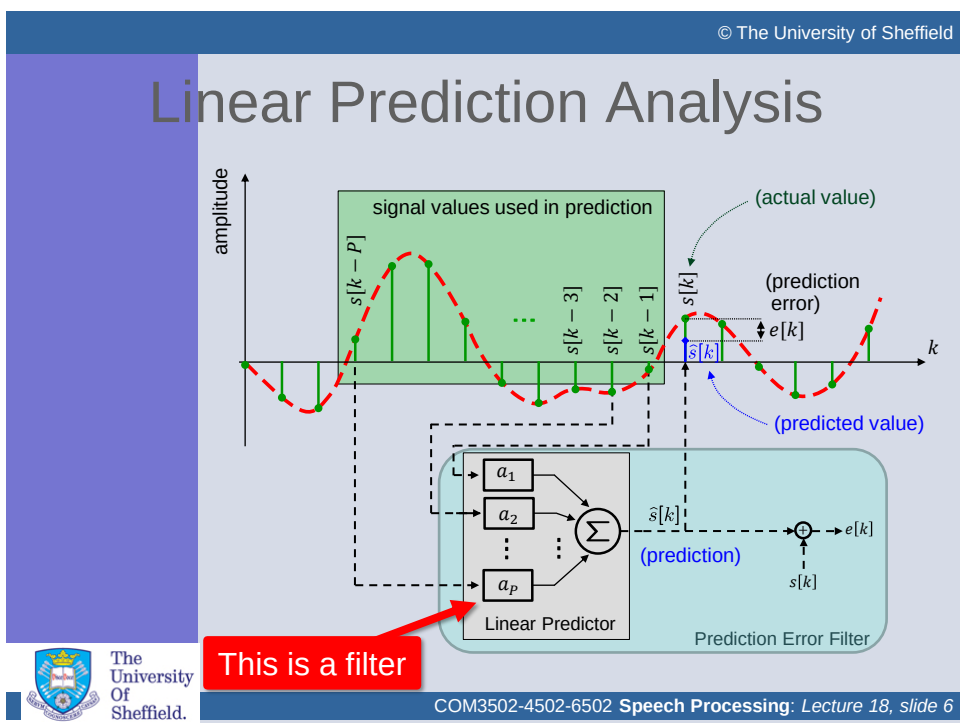
General Digital Filter

Repetition Lecture "Linear Filters"





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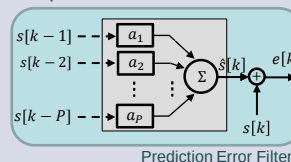


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Linear Prediction Analysis

- The prediction error for a particular sample $e[k]$ is ...

$$\begin{aligned} e[k] &= s[k] - \hat{s}[k] \\ &= s[k] - \sum_{i=1}^P a_i s[k-i] \end{aligned}$$



- Taking Z-transforms of both sides we obtain ...

$$E[z] = \left[1 - \sum_{i=1}^P a_i z^{-i} \right] S[z]$$

- Hence the **prediction error filter** transfer function is ...

$$H_1[z] = \frac{E[z]}{S[z]} = 1 - \sum_{i=1}^P a_i z^{-i}$$

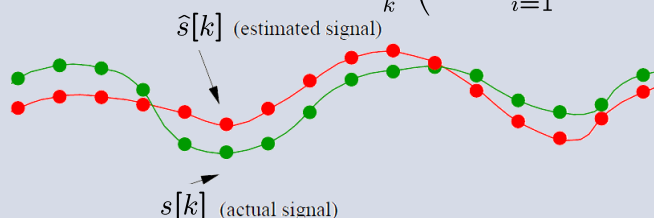
An 'all-zero' filter

Calculating Prediction Coefficients

- The linear prediction coefficients are computed to minimise the sum-squared '**prediction error**' ('**residual signal**') $\bar{e}_p$ over a frame of speech data

$$\begin{aligned} \bar{e}_p &= \sum_k e^2[k] = \sum_k (s[k] - \hat{s}[k])^2 \\ &= \sum_k \left(s[k] - \sum_{i=1}^P a_i s[k-i] \right)^2 \end{aligned}$$

Mean Squared Error (MSE)



- This has the effect of '**whitening**' the error signal (i.e. flattening its spectrum)

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Calculating Prediction Coefficients

Minimum Mean Squared Error (MMSE)

- E_p is minimised by finding the solution to ...

$$\frac{\partial E_p}{\partial a_i} = 0$$
- This results in a set of p simultaneous equations called the 'normal equations' ...

$$\sum_k s[k]s[k-\kappa] = \sum_{i=1}^p a_i \sum_k s[k-i]s[k-\kappa] \quad \text{for } \kappa = 1 \dots p$$
- Different ranges to the summation gives rise to ...
 - the 'autocorrelation method' (Durban's algorithm)
 - the 'covariance method'

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Comparison of Methods

	Autocorrelation	Covariance
Filter order	10-15	10-15
Windowing	✓	✗
Stability	✓	✗
Window size	25 msec	5-10 msec
Computation	+	+++
Reflection coeffs.	✓	✗

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Linear Prediction

- The filter transfer function **from the signal to the prediction error sequence** is ...

$$H_1[z] = \frac{E[z]}{S[z]} = 1 - \sum_{i=1}^p a_i z^{-i}$$

An 'all-zero' filter

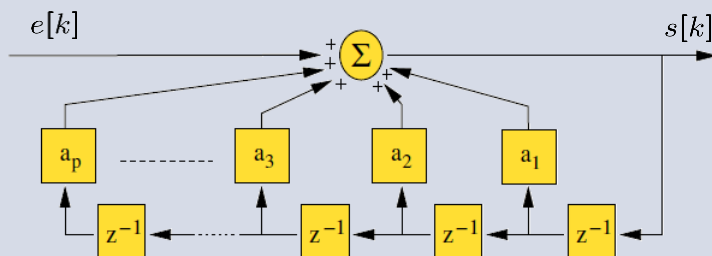
- Hence the filter transfer function **from the prediction error sequence to the signal** is ...

$$H_2[z] = \frac{S[z]}{E[z]} = \frac{1}{1 - \sum_{i=1}^p a_i z^{-i}}$$

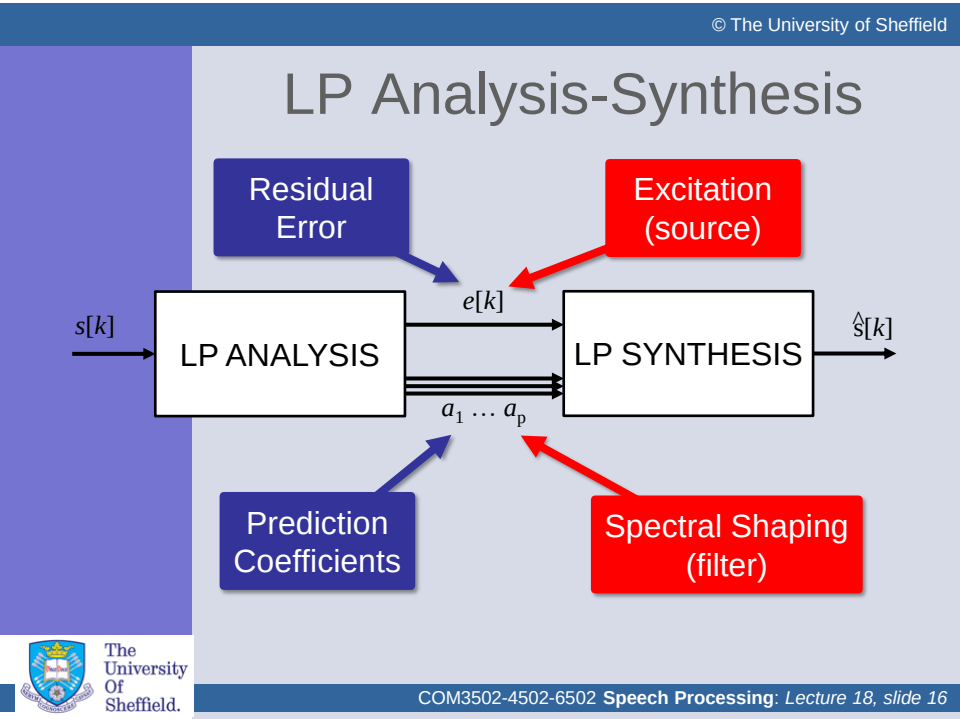
An 'all-pole' filter

Linear Prediction Synthesis

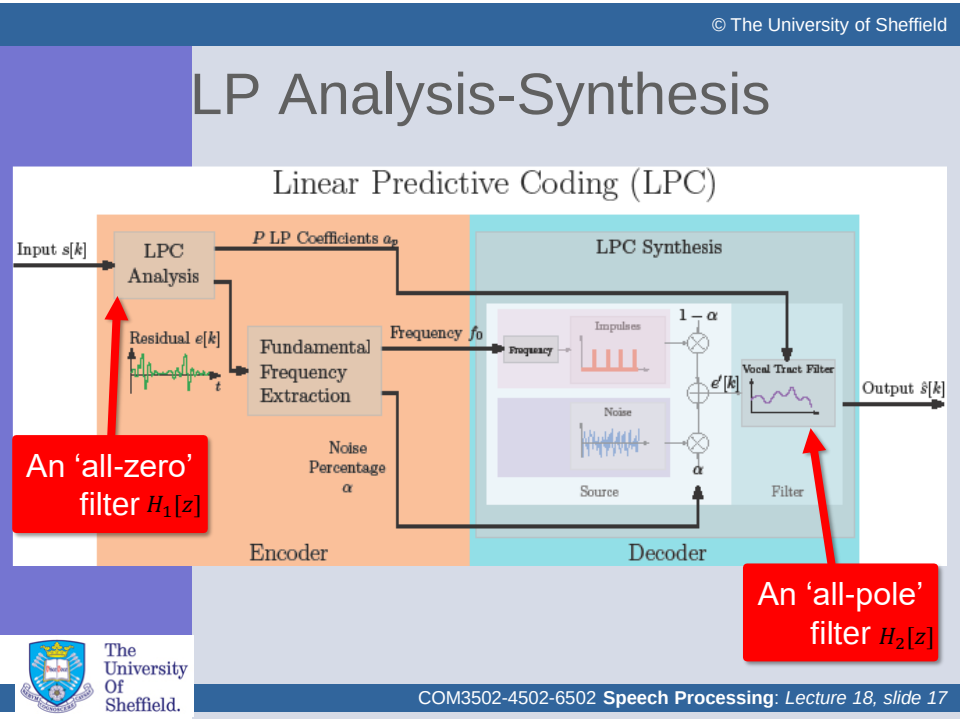
The LP filter can be configured as an **all-pole digital filter** with the error signal as its input ...



$$H_2[z] = \frac{S[z]}{E[z]} = \frac{1}{1 - \sum_{i=1}^p a_i z^{-i}}$$



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LP Analysis-Synthesis

- I.e. the LP filter can be used for analysis and synthesis
 - **analysis**
 - input = signal
 - output 1 = prediction coefficients
 - output 2 = 'residual' error
 - **synthesis**
 - input 1 = prediction coefficients
 - input 2 = 'residual' error
 - output = reconstituted signal
- The data rate of the residual + prediction coefficients can be much lower than the data rate of the signal
- This makes LP particularly suitable for **coding** speech signals



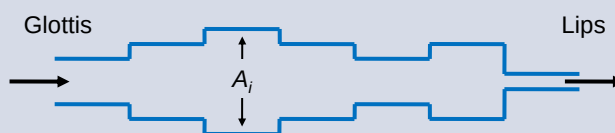
Application to Speech

- In speech analysis-resynthesis ...
 - the LP **filter** models the spectral shaping caused by the vocal tract
 - the error sequence models the **source excitation**
- Note the following assumptions ...
 - the **analysis model** assumes impulse/white-noise excitation
(hence the LP filter also captures spectral shaping caused by the glottal source)
 - the **synthesis model** (all-pole model) ignores any zeros
(such as the nasal side branch and the path back down the glottis)

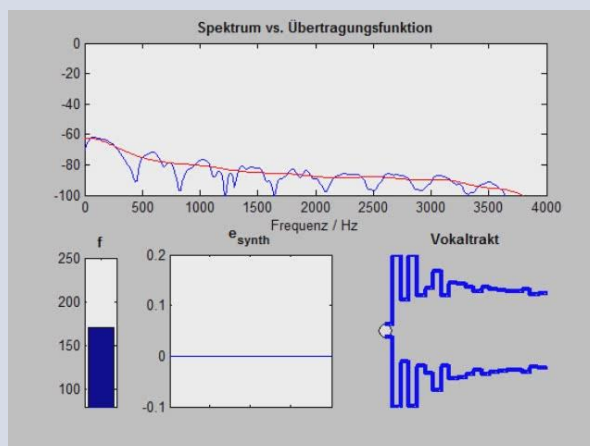


Vocal Tract Area Estimation

- Durban's algorithm (*used in the autocorrelation method*) generates parameters that are known as '**reflection coefficients**'
- It can be shown that these are analogous to the proportion of energy reflected at the boundaries of a lossless acoustic tube model of the vocal tract using a set of cylindrical section of equal length
- The reflection coefficients can thus be used to estimate the cross-sectional area of the vocal tract

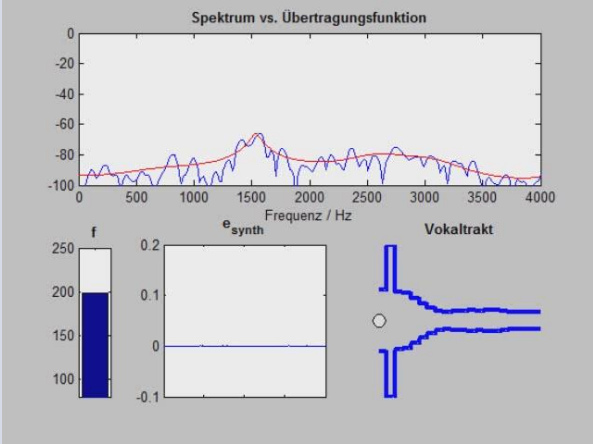


Vocal Tract Area Estimation





Vocal Tract Area Estimation



The plot titled 'Spektrum vs. Übertragungsfunktion' displays the spectrum of a signal (blue line) and its transfer function (red line) across a frequency range from 0 to 4000 Hz. Below the main plot, there are three smaller plots: a bar chart labeled 'f' showing frequency components, a plot labeled 'e_{synth}' showing the synthesized error signal, and a plot labeled 'Vokaltrakt' showing the estimated vocal tract area over time.



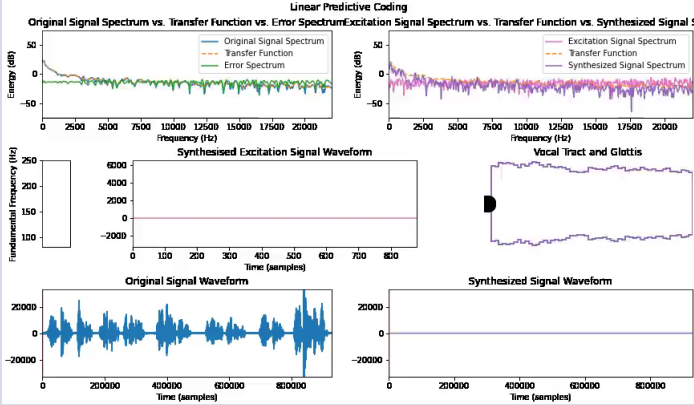
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Vocal Tract Area Estimation



The figure displays several plots related to Linear Predictive Coding (LPC). The top row shows the 'Original Signal Spectrum vs. Transfer Function vs. Error Spectrum' and the 'Excitation Signal Spectrum vs. Transfer Function vs. Synthesized Signal Spectrum'. The middle row shows the 'Synthesised Excitation Signal Waveform' and the 'Vocal Tract and Glottis' area over time. The bottom row shows the 'Original Signal Waveform' and the 'Synthesized Signal Waveform'. The plots show the frequency response and the time-domain waveforms of the original and synthesized signals, as well as the estimated vocal tract area.



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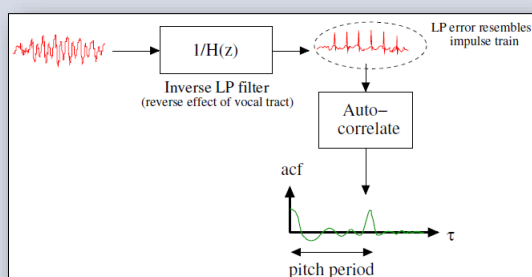
Applications of Linear Prediction

- pitch extraction
- formant analysis
- speech synthesis
- automatic speech recognition
- linear predictive coding (*LPC*)

LP-based Pitch Extraction

- Obtain LP error signal by passing the speech through the *inverse* filter
- Find the pitch frequency by *autocorrelation* on the error signal

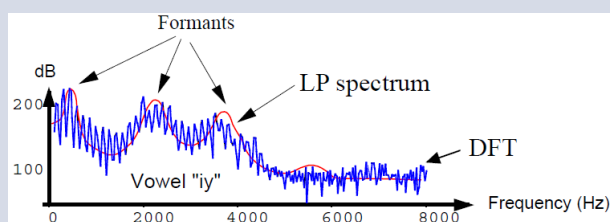
Lab Sheet 7



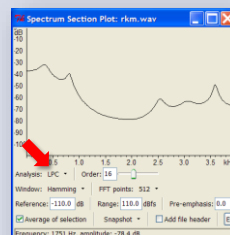
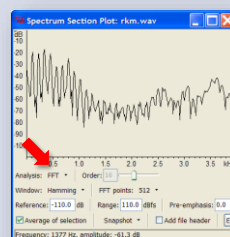
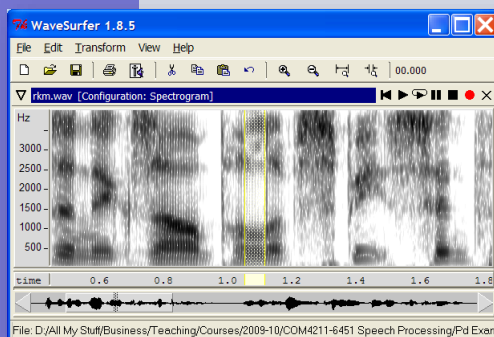
- Use continuity constraints for pitch *tracking*

LP-based Spectrum Estimation

- One way to estimate the speech spectrum is to analyse the frequency response of the LP filter
- Since the excitation information is in the error signal, the LP spectrum is a smoothed all-pole approximation to the speech spectrum
- The LP spectrum is thus useful for calculating spectral features (*such as formant frequencies*)



LP-based Spectrum Estimation



LP-based Formant Extraction

- A pole-pair from LP analysis can model a formant
- There are two main approaches to finding formant candidates ...
 - root-solving from the LP polynomial
 - peak-picking from the LP spectrum
- Root-solving finds the frequencies of the pole-pairs with the smallest bandwidths (*i.e. nearest the unit circle*)
- For high-accuracy formant *tracking* it is necessary to consider alternative candidates

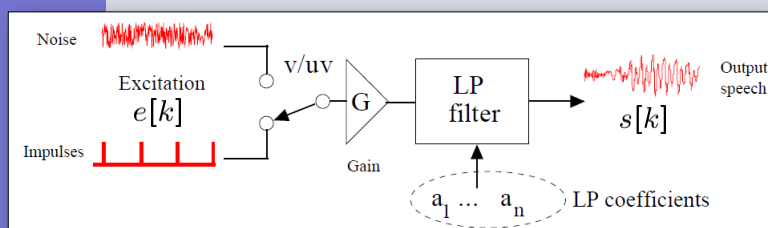


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LP-based Speech Synthesis

An implementation of the *source-filter model*



$$s[k] = G \cdot e[k] - a_1 s[k-1] - \dots - a_p s[k-p]$$



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Linear Predictive Coding (LPC)

- LP *analysis* followed by LP *synthesis*
- Need to quantise and code the excitation and LP filter parameters

The diagram illustrates the LPC process. On the left, the **Transmitter** takes speech samples $s[i]$ and processes them through three parallel blocks: 'Pitch and voicing determination' (outputting f_0 and v/uv), 'Energy estimator' (outputting G), and 'LP analysis' (outputting filter coefficients $a_1, a_2, \dots, a_p$). These outputs are fed into a 'Quantiser and channel encoder' block, which produces a binary-coded excitation signal $c[i]$ (e.g., ...0110100110...). This signal travels through a 'CHANNEL' to the **Receiver**. The receiver's 'Channel decoder' outputs $c[i]$ to an 'Excitation generator' block. This block contains a 'Pulse generator' (driven by f_0 and v/uv) and a 'Noise generator' (driven by G). The outputs of these generators are summed to produce the excitation signal E . This signal E is then passed through an 'LP filter' (using coefficients $a_1, a_2, \dots, a_p$) to produce the 'Reconstructed speech samples' $\hat{s}[i]$.

- Used in NATO 2.4 kbps 'LPC-10e vocoder' 🗣️

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Code-Excited LPC (CELP)

The diagram shows the CELP architecture. A 'Codebook' (a set of stored excitation waveforms) provides input to an 'Excitation Generator' block. The generator's output $e[i]$ is summed with a delayed version of the reconstructed speech signal (via a z^{-T} delay block) to produce the reconstructed speech $\hat{s}[i]$. This signal is then passed through an 'LP filter' to produce the final reconstructed speech samples. An 'Error weighting' block calculates the error between the original speech $s[i]$ and the reconstructed speech $\hat{s}[i]$. This error is fed into an 'Energy' block and an 'Optimiser' block. The 'Optimiser' also receives feedback from the 'Excitation Generator' and the 'LP filter'. It selects the best excitation waveform from the 'Codebook' based on the error and energy, providing an 'Index of excitation waveform' to the 'Channel encoder'. The 'Channel encoder' outputs the index through a 'CHANNEL'.

- Produces good quality speech over 4.8 kbps
- The basis of most modern speech coding systems

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GSM Speech Codec



- The digital mobile cellular radio network which is extensively used throughout Europe and many other parts of the world is termed the '**Global System for Mobile communications**' (GSM)
- The GSM full rate speech codec operates at 13 kbps
- Basic operating principle ...
 - '**CELP**' (Code-Excited LPC)
 - 8 kHz sample rate
 - 20 msec frames
 - 8 short-term predictor coefficients per frame
 - each frame is then further split into four 5 msec sub-frames
 - for each sub-frame the encoder finds a delay and a gain for the codec's long-term predictor
 - the residual signal after both short and long term filtering is quantised for each sub-frame



This slide set has covered ...

- Linear prediction
- LP analysis of speech signals
- Calculating LP coefficients
- Vocal tract area estimation
- LP-based applications
- Linear predictive coding (*LPC*)

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Any Questions ?



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Next time ...

Cepstral Analysis

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